



Darshan
UNIVERSITY

Python Programming - 2301CS404

Lab - 11

Roll No : 459

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Progrmas based on Python Data Structure + UDF + File Handling

Practical 1: Count Lines, Words and Characters

Problem Definition:

Write a Python program to read text from a file and count the total number of lines, words, and characters. The result should be written to an output file.

```
In [2]: fp = open("input1.txt" , "r")

totalChar = len(fp.read())

fp.seek(0)

totalWords = len(fp.read().split())

fp.seek(0)

totalLines = len(fp.readlines())

fp2 = open("output1.txt", "w")

fp2.write("Total Character : " + str(totalChar) + "\n" + "Total Words : " + str(

fp.close()
fp2.close()
```

Practical 2: Search a Word in a File

Problem Definition:

Write a Python program to search a given word in a text file and count its occurrences. Store the result in an output file.

```
In [6]: fp = open("input1.txt" , "r")
fp2 = open("output2.txt","w")

word = input("Enter Word To Search In Input1.txt File :")

fp2.write(word + " : " + str(fp.read().split().count(word)) + " Times")

fp2.close()
fp.close()
```

Enter Word To Search In Input1.txt File : heyy

Practical 3: Student Marks and Average

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Store marks in a list, calculate the average using a user-defined function, and write student details along with average marks to an output file.

```
In [23]: fp = open("marks.txt" , "r")
fp2 = open("output3.txt" , "w")

rollNos = []
names = []
marks = []

def avg(li):
    return str(sum(li)/len(li))

for i in fp:
    rollNos.append(i[0:3])
    names.append(i[4:9])
    marks.append(i[10 : ].split())

print(rollNos)
print(names)
print(marks)

n = len(marks)

ans = []

for i in marks:
    ans.append(list(map(int , i)))

print(ans)

fp.seek(0)
n = len(fp.readlines())
fp.seek(0)
```

```

for i in range(0,n):
    fp2.write("Name : " + names[i] + "\nRollNo : " + str(rollNos[i]) + "\nAvg :

fp.close()
fp2.close()

```

```

['101', '102', '103']
['Vibhu', 'Jay ', 'Mohit']
[['99', '97', '98', '97'], ['99', '99', '98', '97'], ['100', '98', '96', '94']]
[[99, 97, 98, 97], [99, 99, 98, 97], [100, 98, 96, 94]]

```

Practical 4: Maximum and Minimum using Tuple

Problem Definition:

Write a Python program to read numbers from a file, store them in a tuple, and find the maximum and minimum values. Write the result to an output file.

```

In [5]: fp = open("input4.txt","r")
        fp2 = open("output4.txt","w")

        tp = ()

        data = fp.read().split()

        print(data)

        for i in data:
            tp += (int(i),)

        print(tp)

        fp2.write("Max : "+str(max(tp)) + "\nMin : "+str(min(tp)))

        fp.close()
        fp2.close()

```

```

['10', '20', '80', '90', '77']
(10, 20, 80, 90, 77)

```

Practical 5: Remove Duplicate Words using Set

Problem Definition:

Write a Python program to read words from a file and remove duplicate words using a set. Store the unique words in an output file.

```

In [13]: fp = open("input1.txt","r")
        fp2 = open("output5.txt","w")

        # st = set()

        # data = fp.read().split()

        # for i in data:
        #     st.add(i)

```

```

# print(st)

# for i in st:
#     fp2.write(i + " ")

fp.seek(0)

st = set()

for i in fp:
    words = i.split(" ")
    for w in words:
        st.add(w)

print(st)

for i in st:
    fp2.write(i + " ")

fp.close()
fp2.close()

```

```
{'hey\n', 'input', 'file.txt', 'good', 'afternoon', 'ok', 'okk\n', 'by\n', '\n',
'hello', 'its'}
```

Practical 6: Word Frequency using Dictionary

Problem Definition:

Write a Python program to count the frequency of each word in a text file using a dictionary and write the result to an output file.

```

In [17]: fp = open("input1.txt","r")
         fp2 = open("output6.txt","w")

         di = {}

         data = fp.read().split()

         fp.seek(0)

         for i in data:
             di[i] = data.count(i)

         fp2.write(str(di))

         fp.close()
         fp2.close()

```

Practical 7: Student Result and Grade Calculation

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Use a user-defined function to determine grade (A/B/C/Fail) and write the student result to an output file.

Grade Criteria:

- Marks ≥ 75 : Grade A
- Marks ≥ 60 : Grade B
- Marks ≥ 40 : Grade C
- Marks < 40 : Fail

```
In [20]: fp = open("marks.txt" , "r")
fp2 = open("output7.txt" , "w")

def grade(m):
    if m >= 75:
        return "A"
    elif m >= 60:
        return "B"
    elif m >= 40:
        return "C"
    else:
        return "Fail"

rollNos = []
names = []
marks = []

for i in fp:
    rollNos.append(i[0:3])
    names.append(i[4:9])
    marks.append(i[10 : ].split())

print(rollNos)
print(names)
print(marks)

n = len(marks)

ans = []

for i in marks:
    ans.append(list(map(int , i)))

print(ans)

fp.seek(0)
n = len(fp.readlines())
fp.seek(0)

for i in range(0,n):

    fp2.write("Name : " + names[i] + "\nRollNo : " + str(rollNos[i]) + "\nGrade

    for g in ans[i] :
        fp2.write(grade(g) + " ")

    fp2.write("\n\n\n")

fp.close()
fp2.close()
```

```
['101', '102', '103']  
['Vibhu', 'Jay ', 'Mohit']  
[['99', '97', '98', '97'], ['99', '99', '98', '97'], ['100', '98', '96', '94']]  
[[99, 97, 98, 97], [99, 99, 98, 97], [100, 98, 96, 94]]
```